Software Requirements Specification

for

Student Attendance Management System (SAMS)

Version 1.1

Prepared by Team 1

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Revision History

Name	Date	Reason For Changes	Version
Team 1	03-10-2026	Initial version	1.0
Team 1	29-10-2026	Filled traceability matrix after implementation	1.1

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Student Attendance Management System (SAMS) version 1.0. SAMS is a web based application built using Python and Django for recording student attendance and generating attendance reports. This SRS covers the complete system.

1.2 Intended Audience

This document is for:

- Developers - to know what has to be built
- Testers - to write test cases from the requirements
- Faculty evaluator - to review the project
- Admin staff - who will be the actual users of the system

Section 2 gives the overall description, Section 3 the interface requirements, Section 4 the use case diagram, Section 5 the functional requirements and Section 6 the non functional requirements.

1.3 Product Scope

Currently attendance is taken on paper and later entered into excel sheets, which takes time and leads to mistakes in calculating percentages. Students also dont know thier attendance until the shortage list comes out at the end of the semester.

SAMS solves this by:

- letting faculty mark attendance online from a laptop or phone
- calculating attendance percentage automaticaly
- generating reports in PDF and Excel
- letting students check thier attendance anytime

- sending an email when attendance goes below 75%

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- Django Documentation - https://docs.djangoproject.com
- SE Mini Project Guidelines and Problem Statements, Section C

2. Overall Description

2.1 Product Perspective

SAMS is a new, standalone system that replaces the manual attendance register. It is not part of any existing system. It has three tiers:

Tier	Technology
Front end	HTML, CSS, Bootstrap (Django templates)
Back end	Python, Django
Database	MySQL

The system also connects to an external email (SMTP) server to send mails.

2.2 Product Functions

- Login and logout for Admin, Faculty and Student
- Manage students, faculty, subjects and classes
- Mark and update attendance
- View subject wise attendance
- Generate attendance reports and shortage list (PDF / Excel)
- Send shortage email to students below 75%
- Apply for leave and approve / reject leave

2.3 User Classes and Characteristics

User Class	Description	Usage
Admin	College office staff / HOD. Manages students, faculty, subjects and classes. Full access.	Occasional
Faculty	Teachers. Mark attendance, generate reports, approve leave.	Daily

	Most important user class.	
Student	Views own attendance and applies for leave. Mostly uses mobile.	Frequent

2.4 Operating Environment

Component	Requirement
Server OS	Windows 10+ or Ubuntu 20.04+
Runtime	Python 3.10+, Django 4.2
Database	MySQL 8.0
Client	Any modern browser (Chrome, Firefox, Edge) on laptop or mobile

2.5 Design and Implementation Constraints

- Must be developed in Python using Django (as per problem statement)
- MySQL must be used as the database
- Web application only, no mobile app
- Must be completed within the semester by a team of 4
- Code on GitHub, sprints tracked in JIRA

2.6 Assumptions and Dependencies

- College provides student and subject lists in Excel format
- A Gmail SMTP account is available for sending emails
- Faculty have a device with internet during class
- Depends on third party libraries: reportlab (PDF) and openpyxl (Excel)

3. External Interface Requirements

3.1 User Interfaces

- Web interface built with HTML, CSS and Bootstrap
- Role based dashboard shown after login
- Navigation bar with only the options allowed for that role
- Forms show error messages for invalid input
- Attendance page shows the student list with a checkbox for each student

- Pages work on both laptop and mobile screens

3.2 Software Interfaces

Software	Purpose
MySQL 8.0	Stores all data, accessed through Django ORM
Gmail SMTP	Sending shortage and leave emails
reportlab	Generating PDF reports
openpyxl	Reading student Excel uploads and generating Excel reports

3.3 Communications Interfaces

- HTTPS between browser and server
- SMTP on port 587 with TLS for emails

4. Analysis Models

Figure 1 shows the use case diagram with the three actors - Admin, Faculty and Student.

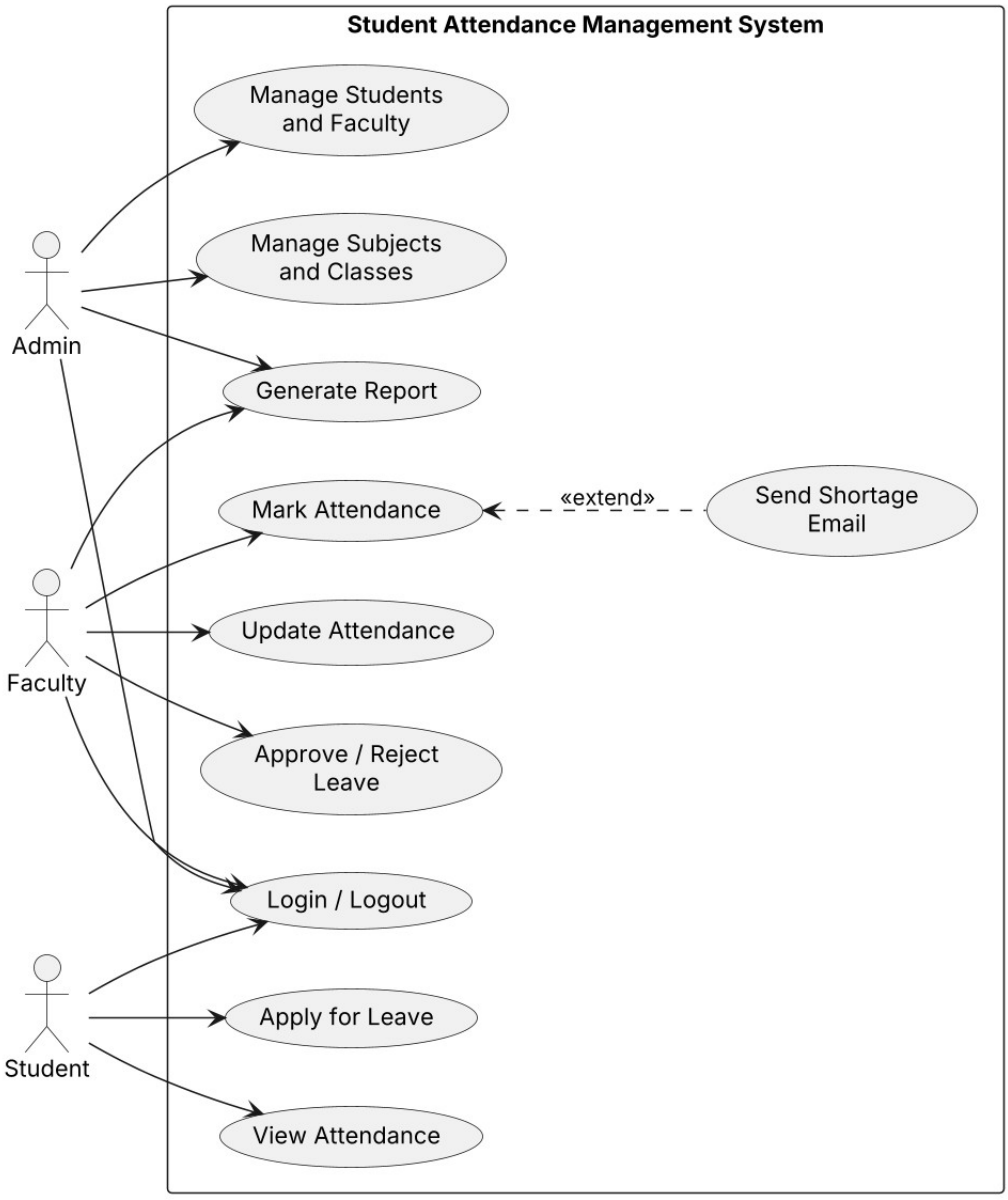


Figure 1: Use case diagram

5. System Features

5.1 User Login and Management

Description and Priority: Lets users login and lets admin manage user accounts. Priority - High.

Stimulus / Response:

User Action (Stimulus)	**System Response**
User enters correct ID and password	Dashboard for that role is shown
User enters wrong password	"Invalid username or password" shown

Admin submits add student form	Student saved, success message shown

Functional Requirements:

ID	Requirement
REQ-1	The system shall allow Admin, Faculty and Students to login using thier ID and password.
REQ-2	The system shall show a different dashboard for each role after login.
REQ-3	The Admin shall be able to add, edit and delete student and faculty accounts.
REQ-4	The Admin shall be able to upload a list of students from an Excel file.

5.2 Subject and Class Management

Description and Priority: Lets admin set up subjects and classes so attendance can be taken. Priority - High.

Stimulus / Response:

User Action (Stimulus)	System Response
Admin adds subject and assigns faculty	Subject saved and shown on faculty dashboard

Functional Requirements:

ID	Requirement
REQ-5	The Admin shall be able to add subjects with subject code, name and semester.
REQ-6	The Admin shall be able to create classes (semester + section) and assign students to them.
REQ-7	The Admin shall be able to assign a faculty to a subject for a class.

5.3 Mark Attendance

Description and Priority: Lets faculty take and correct attendance for thier class. Priority - High.

Stimulus / Response:

User Action (Stimulus)	System Response
Faculty selects subject, class, date and hour	Student list shown
Faculty marks absentees and submits	Attendance saved, confirmation shown

Functional Requirements:

ID	Requirement
REQ-8	The Faculty shall be able to mark each student present or absent for a selected subject, class, date and hour.
REQ-9	The system shall not allow attendance to be marked twice for the same subject, class, date and hour.
REQ-10	The Faculty shall be able to update attendance within 3 days of marking it.
REQ-11	The system shall recalculate attendance percentage of each student after every submission.

5.4 View Attendance and Reports

Description and Priority: Lets students see attendance and faculty/admin generate reports. Priority - High.

Stimulus / Response:

User Action (Stimulus)	System Response
Student opens My Attendance	Subject wise percentage shown
Faculty selects class, subject, dates and clicks Generate	Report created and downloaded

Functional Requirements:

ID	Requirement
REQ-12	The Student shall be able to view thier attendance percentage for each subject.
REQ-13	The Faculty and Admin shall be able to generate an attendance report for a class and subject between two dates.
REQ-14	The report shall be downloadable in PDF and Excel format.
REQ-15	The system shall generate a shortage list of students below 75% attendance.

5.5 Shortage Email and Leave

Description and Priority: Warns students of low attendance and handles leave requests. Priority - Medium.

Stimulus / Response:

User Action (Stimulus)	System Response
Student attendance drops below 75%	Email sent to student
Student submits leave form	Request sent to faculty
Faculty approves leave	Those days marked as Leave

Functional Requirements:

ID	Requirement
REQ-16	The system shall send an email to the student when thier attendance in any subject falls below 75%.
REQ-17	The Student shall be able to apply for leave by giving dates and reason.
REQ-18	The Faculty shall be able to approve or reject a leave request.
REQ-19	Approved leave days shall not be counted as absent.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

ID	Requirement
NFR-1	Every page shall load within 3 seconds with upto 100 users at the same time.
NFR-2	Submitting attendance for a class of 70 students shall take less than 2 seconds.
NFR-3	A full semester report for one class and subject shall be generated within 5 seconds.

6.2 Safety Requirements

ID	Requirement
NFR-4	The database shall be backed up once every day.
NFR-5	Attendance records shall never be permanantly deleted, only updated.

6.3 Security Requirements

Security Objectives:

ID	Objective
SO-1	Only authorised users can access the system and each user can only do what thier role allows.
SO-2	Attendance data can only be changed by the faculty of that subject or the admin.
SO-3	Personal details of students (email, phone) are kept confidential.

Security Requirements:

ID	Requirement	Objective
NFR-6	Passwords shall be stored hashed using Django PBKDF2, never in plain text.	SO-1
NFR-7	The system shall check the user role on every page and block pages not allowed for that role.	SO-1

NFR-8	A student shall only be able to view thier own attendance.	SO-3
NFR-9	User session shall expire after 30 minutes of no activity.	SO-1
NFR-10	All forms shall be protected against CSRF and SQL injection.	SO-2
NFR-11	Every change to an attendance record shall be logged with faculty name, date and time.	SO-2

6.4 Software Quality Attributes

Attribute	**Requirement**
Usability	A new faculty can mark attendance without any training.
Reliability	System available during college hours (8 AM - 5 PM) on working days.
Maintainability	Code split into seperate Django apps per module.
Portability	Runs on both Windows and Linux servers.

6.5 Business Rules

ID	**Rule**
BR-1	Only Admin can add or delete students, faculty and subjects.
BR-2	Only the faculty assigned to a subject can mark attendance for it.
BR-3	Faculty can edit attendance only within 3 days; after that only Admin can.
BR-4	Minimum 75% attendance per subject is needed for ESA eligibility (domain requirement).
BR-5	Attendance % = (classes attended + approved leave) / classes conducted x 100 (domain requirement).

7. Other Requirements

- Student data must be handled as per university data rules.
- Only English language is supported in this version.

Appendix A: Glossary

Term	Meaning
SAMS	Student Attendance Management System
SRN	Student Registration Number
ESA	End Semester Assessment
HOD	Head of Department
ORM	Object Relational Mapping
CSRF	Cross Site Request Forgery
SMTP	Simple Mail Transfer Protocol
Shortage	Attendance below 75% in a subject

Appendix B: Field Layouts

Student Registration Fields

Field	Length	Data Type	Description	Mandatory
SRN	13	Alphanumeric	Student registration number	Y
Name	50	Text	Full name	Y
Email	50	Alphanumeric	College email ID	Y
Phone	10	Numeric	Mobile number	N
Semester	1	Numeric	1 to 8	Y
Section	1	Text	Section	Y
Department	5	Text	e.g. CSE	Y
Password	20	Alphanumeric	Changed on first login	Y

Report Fields

Attendance Report	**Shortage Report**
Subject Code and Name	Subject Code and Name
Class (Sem and Section)	Class (Sem and Section)
From Date / To Date	SRN
SRN	Student Name
Student Name	Classes Conducted
Classes Conducted	Classes Attended
Classes Attended	Attendance %
Leave Days	Email Sent (Y/N)
Attendance %	Faculty Name
Faculty Name	Generated On
Generated On	

Appendix C: Requirement Traceability Matrix

Sl. No	**Req ID**	**Brief Description**	**Architecture Ref**	**Design Ref**	**Code File Ref**	**Test Case ID**	**System Test Case ID**
1	REQ-1	allow Admin, Faculty and Students to login using thier ID and password	accounts	SAD 4.1	accounts/ views.py	LoginTests	TC-01, 02
2	REQ-2	show a different dashboard for each role after login	accounts	SAD 3.9	accounts/ decorators.py	DashboardTests	TC-03
3	REQ-3	Admin can add, edit and delete student and faculty accounts	accounts	SAD 3.4	accounts/ admin.py	-	TC-05
4	REQ-4	Admin can upload a list of students from an Excel file	accounts	SAD 4.3	accounts/ importer.py	StudentExcelUploadTests	TC-04

5	REQ-5	Admin can add subjects with subject code, name and semester	academics	SAD 3.6	academics/ admin.py	AcademicsConstraintTests	TC-05
6	REQ-6	Admin can create classes (semester + section) and assign students to them	academics	SAD 3.6	academics/ admin.py	AcademicsConstraintTests	TC-05
7	REQ-7	Admin can assign a faculty to a subject for a class	academics	SAD 3.6	academics/ models.py	AcademicsConstraintTests	TC-05
8	REQ-8	Faculty can mark each student present or absent for a selected subject, class, date and hour	attendance	SAD Fig 2	attendance/ views.py	MarkAttendanceTests	TC-06
9	REQ-9	not allow attendance to be marked twice for the same subject, class, date and hour	attendance	SAD Fig 2	attendance/ views.py	MarkAttendanceTests	TC-07
10	REQ-10	Faculty can update attendance within 3 days of marking it	attendance	SAD 4.3	attendance/ views.py	EditAttendanceTests	TC-08
11	REQ-11	recalculate attendance percentage of each student after every submission	attendance	SAD 4.1	attendance/ utils.py	AttendancePercentTests	TC-09
12	REQ-12	Student can view thier attendance percentage for each subject	reports	SAD 4.3	reports/views.py	-	TC-10
13	REQ-13	Faculty/Admin can generate an attendance report for a class and subject between two dates	reports	SAD 4.3	reports/ services.py	ClassReportTests	TC-11
14	REQ-14	Report downloadable in PDF and Excel format	reports	SAD 4.3	-	-	TC-11
15	REQ-15	generate a shortage list of students below 75% attendance	reports	SAD 4.3	reports/views.py	ShortageListTests	TC-12
16	REQ-16	send an email to the student when thier attendance in any subject falls below 75%	leaves	SAD Fig 2	leaves/utils.py	ShortageEmailTests	TC-13
17	REQ-17	Student can apply for leave	leaves	SAD Fig	leaves/views.py	ApplyLeaveTest	TC-14

		by giving dates and reason		3		s	
18	REQ-18	Faculty can approve or reject a leave request	leaves	SAD Fig 3	leaves/views.py	ApproveLeaveTests	TC-14
19	REQ-19	Approved leave days shall not be counted as absent	leaves	SAD Fig 3	leaves/views.py	ApproveLeaveTests	TC-14